Betula alleghaniensis Britton

Yellow Birch

Betulaceae -- Birch family

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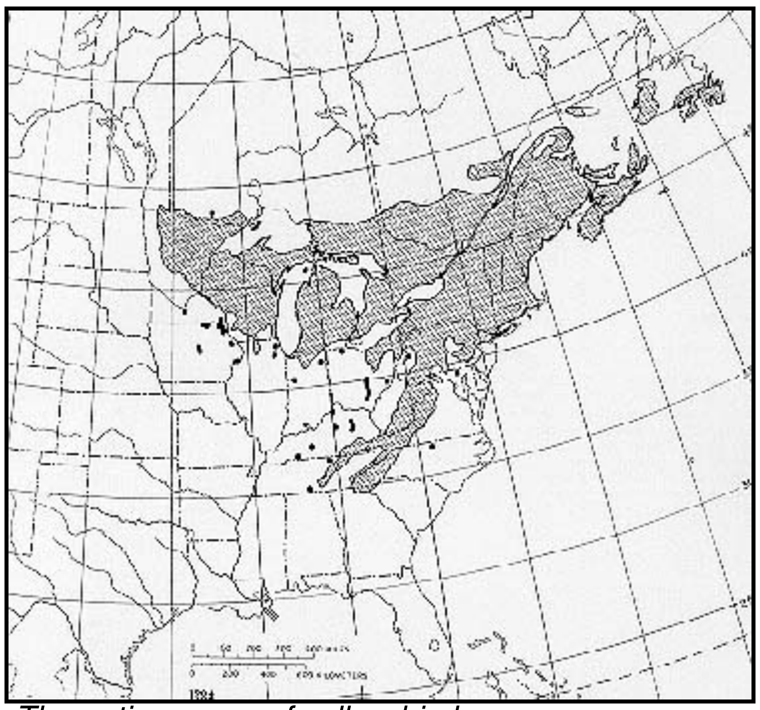
Yellow birch *(Betula alleghaniensis) is* the most valuable of the native birches. It is easily recognized by the yellowish-bronze exfoliating
and has a flavor of wintergreen. Other names are gray birch, silver birch, and swamp birch. This slow-growing long-lived tree is found v
soils of the uplands and mountain ravines. It is an important source of hardwood lumber and a good browse plant for deer and moose.

Habitat

Native Range

Yellow birch ranges from Newfoundland, Nova Scotia, New Brunswick, and Anticosti Island west through southern Ontario to extreme s
northeastern Iowa; east to northern Illinois, Ohio, Pennsylvania to northern New Jersey and New England; and south in the Appalachia
Georgia. Southward yellow birch grows at higher elevations, appears more sporadically, and finally is restricted to moist gorges above

The largest concentrations of yellow birch timber are found in Quebec, Ontario, Maine, Upper Michigan, New York, and New Brunswi
yellow birch in North America is in Quebec.



-The native range of yellow birch.

Climate

Yellow birch grows in cool areas with abundant precipitation. Its northern limit coincides with the 2° C (35° F) average annual temperatu
with the 30° C (86° F) maximum temperature isotherm (31). Although the average annual temperature is about 7° C (45° F) throughout
38° C (-40° F to 100° F) (45). Annual precipitation rtinges from about 1270 mm (50 in) in the East to 640 mm (25 in) in central Minneso
may be snow. Snowfall ranges from 152 to 356 cm (60 to 140 in) and averages 229 cm (90 in) in the north. The growing season ranges

Soils and Topography

Yellow birch grows over a large area with diverse geology, topography, and soil and moisture conditions. In Michigan and Wisconsin it i

deposits, shallow loess deposits, and residual soils derived from sandstone, limestone, and igneous and metamorphic rock (95). Soils
other parts of its range.

Growth of yellow birch is affected by soil texture, drainage, rooting depth, stone content in the rooting zone, elevation, aspect, and fert
and moderately well-drained sandy loams within the soil orders Spodosols and Inceptisols and on flats and lower slopes (45). It also
region. Rootlet development is profuse in loam but poor in sand. Even though its growth is poor, yellow birch is often abundant whe
species is less severe.

In the Lake States birch grows best on well- and moderately well-drained soils and on lacustrine soils capped with loess. Its growth is
limestone, and coarse-textured sandy loams without profile development (95). Site quality between the best and poorest sites differs b

In the Green Mountains of Vermont birch grows on unstratified glacial till up to 792 m (2,600 ft) (109). Here, thickness of the upper soi
used to estimate site index-birch grows better at lower elevations than higher elevations and on northeast aspects than southwest asp

Associated Forest Cover

Yellow birch is present in all stages of forest succession. Second-growth stands contain about the same proportion (12 percent) of birc
in small pure groups in mixtures with other species. Because yellow birch is seldom found in pure stands, it is not recognized as a sep

Yellow birch is a major component of three forest cover types: Hemlock-Yellow Birch (Society of American Foresters Type 24), Suga
Yellow Birch (Type 30) (41). Hemlock-Yellow Birch is considered a long-lasting subclimax type, as is Red Spruce-Yellow Birch, except

Yellow birch is a commonly associated species in the following forest cover types (41): Balsam Fir (Type 5), Pin Cherry (Type 17),
White Pine-PineNorthernNorthern Red Oak-Red Maple (Type 20), Eastern White Pine (Type 21), White Pine-Hemlock (Type 22), E
26), Sugar Maple (Type 27), Black Cherry (Type 28), Red Spruce-Sugar Maple-Beech (Type 31), Red Spruce (Type 32), Red Spruc
Paper Birch-Red Spruce-Balsam Fir (Type 35), Northern White-Cedar (Type 37), Black Ash-American Elm-Red Maple (Type 39),
(Type 58), Yellow-Poplar-White Oak-Northern Red Oak (Type 59), Beech-Sugar Maple (Type 60), White Spruce (Type 107), and Red

Yellow birch commonly grows in stands with sweet birch *(Betula lenta),* eastern hophornbeam *(Ostrya virginiana),* American hornb
Aspen (Type 16). Small trees and shrubs commonly associated with yellow birch in the north are striped maple *(Acer pensylvanicu*
(Cornus alternifolia), beaked hazel *(Corylus cornuta),* Atlantic leatherwood *(Dirca palustris),* witch-hazel *(Hamamelis virginiana),* fly
ash *(Sorbus americana),* American elder *(Sambucus canadensis),* Canada yew (Taxus *canadensis),* mapleleaf viburnum *(Viburnum*
deep, narrow gorges and upper coves of the southern Appalachians, yellow birch is frequently associated with three evergreen
(Leucothoe catesbari), and rosebay rhododendron *(Rhododendron maximum)*and with striped maple, sassafras *(Sassafras albidum),*

Life History

Reproduction and Early Growth

Flowering and Fruiting- Yellow birch is monoecious; male and female catkins are borne separately on the same branch. Erect stamin
two or three, or singly at the tips of long shoots. Individual florets of a staminate catkin contain three flowers and each flower has three
(2.2 in) long' remain on the tree over winter. In the spring they turn purplish yellow while elorigating to 7 to 10 cm (2.8 to 3.9 in) and the
also formed during the fall are enclosed in buds over winter; they appear terminally as greenish catkins (each about 2 cm or 0.8 in) on
Each floret contains three naked flowers and is subtended by a three-lobed bract. Each flower consists of a bicarpellary ovary with two
before pollen is shed on the same tree (29). Normally only one of the two ovules in each chamber develops into a seed; the other thre
fertilized and two seedlings emerge from one seed (17).

The fruit, a winged nutlet, ripens in late August or early September. Each nutlet averages from 3.2 to 3.5 mm (0. 13 to 0. 14 in) long
seed coats and firm white interiors.

Seed Production and Dissemination- Normally, 40 years is considered the minimum and 70 years the optimum seed-bearing age fo
produced by 30- to 40-year-old trees in either open-grown positions or thinned stands. However, open-grown progeny test saplings wi
catkins at 8 years (21). Germinable seed has also been obtained from 16-year-old trees 5 to 10 cm (2 to 4 in) in d.b.h. and 6.7 to 7.6 r

Good seed crops usually occur at about 2 to 3 year intervals but the frequency of good or better seed crops varies-every 1 to 4 years
Maine, and every 3 years in Ontario (10). Consecutive good or better seed crops only occurred once in the 26-year Wisconsin stud
Seed-crop failures are often caused by hard frost in late spring or early fall or by insects and disease. The percentage of viable seed
high proportion of seedcoats without embryos, probably caused by parthenocarpy (14). Seed viability is often affected by wea
development. It also varies by locality, stand, and individual trees within the same stand.

Although some seeds fall shortly after they mature in August, the first heavy seedfall in Canada and the northern United States comes larger seeds are not shed first nor is their germination capacity any better than that of smaller filled seeds (17).

Yellow birch seeds are light, averaging 99,200/kg (45,000/lb) (8). They are dispersed by the wind and blown up to 400 m (1,320 ft) ov regeneration is at least 100 m (330 ft) from the edge of a fully stocked mature northern hardwood stand.

Yellow birch is a prolific seeder, producing between 2.5 and 12.4 million seeds per hectare (1 to 5 million/acre) in good seed years (seed year (52). Seed viability is usually good in years with heavy seed crops and poor in years with light crops. Germinative capacity 20 percent under natural conditions.

The next seed crop can be estimated from the abundance of the overwintering male catkins (88). Fairly reliable estimates of the fall ye the spring-maturing red maple crop (47).

Yellow birch seedcoats contain a water-soluble germination inhibitor that is inactivated by light (17). Seed dormancy can be broken do sand at 5° C (41° F) for 4 to 8 weeks or by germinating unchilled seeds in a water medium under "cool-white" fluorescent light for more when unchilled rather than stratified seeds are used (8). Following stratification, seeds are germinated at alternating day and night ten 40 days, and alternating temperature of 30° C and 20° C (86° F and 68° F) with at least 8-hour light periods are used for unchilled se common in good seed years.

Seeds can be stored in tightly closed bottles at from 2 to 4° C (36 to 40° F) for 4 years without losing viability (17). Some seed lot germination after 12 years (19).

Seedling Development- Yellow birch seeds dispersed in the fall and winter germinate at warm temperatures in early June. Germinatio regenerate on mossy logs, decayed wood, rotten stumps, cracks in boulders, and windthrown hummocks because hardwood leaf litter seeds germinate in compacted leaf litter that birch radicals and hypocotyls cannot pierce (10). Drying of the litter during the growing se succumb to frost damage or are smothered by the next leaf fall.

Unless stands have been burned or heavily disturbed by blowdowns or logging, abundant birch regeneration is normally restricted to On less-well-drained soils, sufficient moisture remains in the leaf litter to result in adequate establishment if advance regeneration of ot

The most important factors affecting the catch of yellow birch seedlings are an adequate seed supply, favorable weather, proper seedb

Removing advance regeneration is at least as important as preparing proper seedbeds (121). Scarification fulfills both requirements a increase the initial catch of birch seedlings (45). Optimum seedling survival and growth, however, occur on disturbed humus or mixed regeneration (126).

Mechanical scarification and prescribed burning are used to prepare receptive seedbeds and eliminate advance regeneration. Scari and to expose 50 to 75 percent of the area (46,88). Spring burning during and shortly after leafout in years with abundant male b regeneration and provide seed for successful birch regeneration. Treatments should coincide with good seed crops because the effect seasons.

Under dense forest canopies (13 and 15 percent of full sunlight) yellow birch roots grow slower than sugar maple seedlings (81). selection cutting (43).

The optimum light level for top growth and root development of birch seedlings up to 5 years old is 45 to 50 percent of full sunligh similar light levels (86). In field studies, the greatest 2-year height growth occurred at the lowest canopy density of 0 to 14 percent, ar under canopy densities between 29 and 50 percent (123). Moderate side shade is beneficial to birch seedlings during their first 5 years

Clearcutting small patches or strips provides suitable conditions for yellow birch seedling establishment in the Northeast where rainfall (0.1 to 0.6 acre) produce good catches of birch regeneration. Patches are difficult to manage but can be used in uneven-aged manage of mature or defective trees are harvested (85).

In the dry western part of the species range, success with strip clearcutting to regenerate birch has been too variable to generally reco good seed crop, favorable weather, and control of advance regeneration (89). Although strips 20 and 40 in (66 and 132 ft) wide were e in Michigan, strips 20 in (66 ft) wide are about optimum in Canada (10), and strips 15 in (50 ft) wide are recommended in the Northeast

Clearcuttings of 2 to 4 ha (5 to 10 acres) and uniform selection cuttings are not as effective as smaller patches or the shelterwood m western part of its range, birch regenerates best under shelterwood cuttings (48,121). Ten well-distributed yellow birch seed trees pe Otherwise, 0.56 kg/ha (0.5 lb/acre) of stratified (6 to 8 weeks at 5' C (41' C) birch seed can be applied about a week after site prepar January (48).

Yellow birch can also be successfully established by planting 2-0 stock 15 to 50 cm (6 to 20 in) tall on 0.08 ha (0.2 acre) clearcut patch

Yellow birch seedling growth in the Northeast is limited by inadequate soil fertility in acid sandy subsoils and can be greatly improve phosphorus deficiency and aluminum toxicity (61). Aluminum is toxic to roots, especially in subsoils, deficient in magnesium and sulfu or up to 80 p/m but concentration of 120 p/m or more are toxic (84).

Manganese toxicity in seedlings occurs above foliar concentrations of more than 1,300 p/m; concentrations of less than 60 p/m are def

Optimum nursery seedbed density is about 160 seedlings per square meter (15/ft²) (45). Normally 2-0 stock averaging 28 cm (11 in) t caliper is large enough for dormant spring planting. For early starts in the greenhouse, seedlings require at least 2 months of cold s feasible (11,50,115). In 3 months seedlings 40 to 50 cm (16 to 20 in) tall can be produced in the greenhouse using 20-hour days with s (17). Growth can also be accelerated by using plastic greenhouses (94).

After 5 years, yellow birch seedlings are normally overtopped by faster-growing species and require complete release from overstory s Photosynthetic rates of overtopped seedlings are only 54 to 70 percent of those grown in full sunlight and their dry weights are 66 perc

Birch crop trees in Vermont and Michigan seedling stands (up to 2.5 cm or 1 in d.b.h.) have benefited from cleaning or early release (4 radius of the bole radius in Michigan exhibited the best stem, crown, and branch characteristics. They averaged 2 cm (0.8 in) large Shoot growth of yellow birch partly depends upon current photosynthate (73). The shoot elongation period for released saplings (1.5-m more light and moisture available to them (55).

Vegetative Reproduction- Yellow birch seedlings and small saplings reproduce from sprouts when cut, but sprouting from larger stem

Greenwood cuttings of birch have been successfully rooted (45) and overwintered (56). The species can also be propagated by graftin

Sapling and Pole Stages to Maturity

Growth and Yield- Yellow birch requires overhead light, crown expansion space, soil moisture, and nutrients to compete with its faster (39,54,55,57), poles (36), and small saw logs (37) in the Lake States and the Northeast demonstrate that 16- to 65-year-old trees respo favorable growing positions throughout their lives. Growth rates, however, gradually decline as trees age.

In the sapling stage, growth rates can be increased up to 8 cm (3 in) per decade by release of dominant and codominant crop tree crow to 8 ft) of the crop trees' crown perimeters (39).

Diameter growth rates of pole-size trees can be increased from 75 to 78 percent by providing the same open growing space be codominant crop trees with well-developed crowns respond best to release. Complete crown release provides adequate crown and ro development in yellow birch. In practice no more than 247 well-spaced crop trees (6.4 rn or 21 ft apart) per hectare (100/acre) are re d.b.h. per hectare (61/acre).

Diameter growth rates of saw log-size trees can also be increased by about 45 percent by either removing two important crown comp (37). Through careful tending with even-aged management techniques, trees 46 cm (18 in) in d.b.h. can be produced in less than 90 ye

Even-aged and uneven-aged stocking guides for northern hardwoods have been developed for mixed stands in the Lake States (40 these guides suggest leaving a residual stand of 60 to 80 percent of full stocking and retaining increasing amounts of basal areas as the amounts of birch saw log volumes are obtained from selectively cut stands. Growth rates and yields from mixed softwood stands a improvement should be left after thinning or crown release. Yellow birch trees are financially mature at 56 cm (22 in) in d.b.h. but t defects prevent improvement to top-grade saw or veneer logs (78).

Yellow birch prunes itself well as long as its crown is allowed to close within 5 or 6 years after release. It can, however, be pruned to should be done on small, fast-growing trees with small knotty cores to limit discoloration and keep decay organisms from entering wou pruned flush without causing lumber defects. Most wounds up to 5 cm (2 in) close within 7 years (112).

Yellow birch trees are sensitive to excessive exposure following heavy cutting and commonly develop epicormic branches from dorm more than dominant and codominant trees. Epicormic sprouting increases with intensity of release but is not a serious problem in closure (36). Sprouting is usually more profuse just beneath the live crown than further down the stem.

In 60- and 80-year-old New Hampshire stands, addition of nitrogen, phosphorus, calcium, and magnesium to the acid subsoil of Sp suggested for alleviating these deficiencies in yellow birch and for correcting aluminum toxicity in the roots and high manganese economical way to correct calcium deficiency of stems and leaves and aluminum toxicity of roots (65).

Six percent and 51 percent increases in average 8-year basal area growth of 90-year-old trees have been reported from applic potassium, respectively (98). Lime increased crown-sectional area growth of 70-year-old thinned trees in New York 1 year after broadc available for uptake (77).

Lake States fertilizer studies in a 65-year-old sawtimber stand and two pole-size stands showed no significant diameter growth resp phosphorus, and potassium broadcast fertilizer applications that occurred within 3 years of treatment (37,116).

Yellow birch is one of the slowest growing components of both old growth and unmanaged second growth northern hardwood forests are common in unmanaged stands or stands managed under the uneven-aged system. Fifteen-year mortality exceeded ingrowth fo (35).

Complete tree and component part biomass equations have been published for yellow birch trees from 0.25 to 66 cm (0.1 to 26 in) ir (128) and in New Brunswick (69).

Site index curves from stem analysis data have been developed for yellow birch growing in northern hardwood stands in northern Wisc

The Vermont curves, when corrected, are also applicable in New Hampshire (110). In northern Wisconsin and Upper Michigan yellow b birch, sugar maple, and red maple have similar site indexes up to age 50 on well-drained soils. On less well-drained soils, yellow birch States and New England, average site index is about 16.8 to 19.8 m (55 to 65 ft) for birch at age 50. Until age 50 height growth is fas these height growth patterns are reversed.

In New England mature trees on medium sites have attained a height of up to 30.5 m (100 ft) and a d.b.h. of more than 76 cm (30 in) a years in unmanaged forests (9). The largest specimen on record, near Gould City, MI, is 144 cm (56.7 in) in d.b.h. and 34.7 m (114 ft) near Big Bay, MI, is 151 cm (59.5 in) in d.b.h. and 32.6 m (107 ft) tall with a 26.2 m (86 ft) crown spread (58). Yellow birch trees are co ages of more than 366 years.

Rooting Habit- Yellow birch has an adaptable well-developed, extensive lateral root system. Its roots are capable of either spreading h of more than 1.5 rn (5 ft) under favorable conditions. Roots often follow old root channels in compacted soil layers. Rooting patterns of their origin on decayed wood and stumps (fig. 3). Within- and between-tree root grafting is common in birch (42).

Irregularly distributed lateral roots of sapling and pole-size trees often extend well beyond their crown perimeters (120). Most root syst on slopes are usually concentrated along the contour and the uphill side of the stem. Main laterals are close to the soil surface and usu stem. These sinkers often penetrate to impervious layers (53). Replacement root growth is active from leafout (May 5) until late Octobe

Reaction to Competition- Yellow birch is generally considered intermediate in shade tolerance and competitive ability (45). It is more tolerant than its major associates, sugar maple *(Acer saccharum),* beech *(Fagus grandifolia),* and hemlock *(Tsuga).* Yellow birch is the Sugar Maple- Beech-Yellow Birch and Hemlock-Yellow Birch. It cannot regenerate under a closed canopy; it must have soil disturbance stable on moist sites but gives way with age to more tolerant species on dry sites.

Yellow birch is often a pioneer species following fires but is usually less abundant than aspen *(Populus)*, pin cherry *(Prunus pensylva* cannot compete successfully with advance regeneration, grass, and herbaceous plants. An allelopathic relation between yellow b Advance sugar maple regeneration offers the stiffest competition in the Sugar Maple-Beech-Yellow Birch cover type, while red maple wetter sites in the Lake States.

Damaging Agents- Yellow birch is a very sensitive species that is more susceptible to injury than its common associates. It is windfirm subject to windthrow on shallow, somewhat poorly drained soils. Thin-barked yellow birch is susceptible to fire injury. Seedlings and sa fine branching habit of yellow birch makes it susceptible to damage from accumulating ice or snow loads. Large trees are frost hardy (9 especially on litter seedbeds under full and partial shade. Winter sunscald can be a problem on the south and southwest sides of birch salt spray. Seed germination is also greatly reduced by 0.20-percent salt concentrations in the soil (6). Simulated acid rain at pH values foliar damage occurs at pH levels of 3 or less and seedling growth reductions at pH 2.3 (127). Yellow birch seedlings are tolerant to atm sensitive to 3.5 p/m of sulfur dioxide (66).

Post-logging decadence is a localized decline from which most trees recover. It consists of top dying and some mortality following hea is more susceptible to root, stem, or crown injury and more severely affected than its common hardwood associates. Weakened tre birch borer.

A decline of yellow birch and paper birch trees, called birch dieback, caused widespread mortality between 1932 and 1955 in eastern C birches of all sizes, even in undisturbed virgin stands. The first visible symptoms of dieback are similar to those of decadence. Foliage yellowish, and thin. Following this, tips of branches die, then dying progresses downward, involving entire branches and often more tha killed within 3 to 5 years by the bronze birch borer, which with root rot fungi, the gold ring spot virus, and other pests have been conside Many researchers have attributed birch dieback to adverse climatic conditions, drought, and increased soil temperature, over an exten the trees and predisposed them to attacks by the borer. Others have considered over-maturity, past cutting practices, killing of associat defoliating insect outbreaks on birch as initially responsible for weakening the trees. More recently the apple mosaic virus (49) and the possible triggering mechanisms for birch dieback. Under the "frozen soil" theory, shallow-rooted birch trees in years without snow cove their stems through both frozen rootlets and those broken from frost heaving. To date, no single "triggering" cause of birch dieback has result of one or more of the indicated stress factors.

Top-dying and reduced growth of yellow birch crowns have also been associated with heavy birch seed crops (51). This dieback occu peripheral 0.6 to 0.9 m (2 to 3 ft) of branch tips on mature trees and usually just the past season's growth on younger trees.

Discoloration and decay are the major causes of defect and loss in wood quality of yellow birch (75,92). Discoloration and decay deve northern hardwood species (60,107). Some of the nonhymenomycetes most frequently isolated from discolored wood associated *canadense, Phialophora spp., Phialocephala spp., Hypoxylon spp.,* and *Nectria* spp. (32,103,104,107).

Mechanical wounds with more than 320 cm² (50 in²) of exposed wood are important entrance courts for decay fungi (92). *Pholi concentrica,* and *Hypoxylon spp.* are aggressive invaders of these larger wounds (76,104,106). *D. concentrica* and *Hypoxylon sp* associated with larger mechanical injuries more than 20 years old and frost cracks more than 10 years old. Species of *Phialophora a Graphium spp., Phialophora spp., Polyporus spp., Pholiota spp.,* and *Nectria* are the microorganisms most frequently associated w wounds cause reddish-brown decay columns from 74 to 213 cm (29 to 84 in) long within 2 years following boring.

Nectria galligena is the most common and damaging stem disease of yellow birch. It causes perennial targetlike cankers, a twig blight, penetrate saplings, small branches, buds, and wounds but usually enters the host through cracks originating at branch axils from heav defects that reduce stem quality and weaken the stem, increasing the chances for wind breakage (45).

Diaporthe alleghaniensis causes a black sunken canker and shoot blight of yellow birch (1). Natural infections probably enter through b other injuries (2). Cankers appear on shoots, stems, and petioles of seedlings in the spring and summer and foliage wilts and browns i when conditions are optimum for infection and growth (70). Normally the fungus is weakly pathogenic and thins out less vigorous and c

Gnomonia setacea causes a canker, shoot blight, and leaf spot disease of yellow birch seedlings (71).

Stereum murrayi causes elongated, sunken, bark-covered stem cankers and a yellow-brown stringy trunk rot of yellow birch. Cankers a about 0.3 m (1 ft) above and below cankers on pole-sized trees (106). Decay can be extensive in overmature yellow birch. *Phellinus la* covered cankers on mature and over-mature trees. Single cankers indicate extensive decay. It is more common on dead than living tre conks that develop in trunk wounds and branch stubs. Sometimes conks of *L. obliquus* and *Phellinus igniarius* occur on dead branch s indicates from 50 to 100 percent cull (59) and decay extends from 1.5 to 2.1 m (5 to 7 ft) above and below each conk. *Inonotus obliqu* tissues around these sterile conks (106).

Armillaria mellea, the shoestring root rot, is the most common and important root and butt decayer of yellow birch trees (106). The fung roots.

Inonotus obliquus, Pholiota spp., *Phellinus igniarius,* and *P. laevigata* are the principal decay fungi of yellow birch trunks (45,76,106). T white trunk rot of yellow birch. A single conk indicates extensive decay that extends 2.4 to 3.0 m (8 to 10 ft) above and below the conk. larger birches and *Pholiota limonella* causes a yellow-brown stringy trunk rot.

Ganoderma applanatum usually occurs on dead birches but sometimes rots the centers of trunks and infects roots and butts through w *fomentarius,* the tinder fungus, are common on dead birch. The fungus also has been associated with decay in living and dead branch *pinicola,* and *Polyporus lucidus* also are primarily decayers of dead wood but they may extend into centers of living trees (59).

Coniothyrium spp., common twig-inhabiting fungi, injure yellow birch seeds and seedlings (108). They are associated with weevils that

The bronze birch borer *(Agrilus anxius) is* the most serious insect pest of yellow birch. It attacks both healthy and weakened birches (8 in dead or dying wood in weakened trees. Mature and overmature trees left severely exposed after logging and in lightly stocked sta stands. Adults deposit their eggs in bark crevices of upper branches. Grubs hatch, bore meandering tunnels underneath the bark that stem and kill the tree within 2 or 3 years. The Columbian timber beetle *(Corthylus columbianus)* bores deep into the sapwood *(Chrysobothris sexsignata)* occurs commonly on birch in the East. The ambrosia beetle *(Xyloterinus politus) is* a secondary insect t holes through lenticles in the bark and make galleries (105).

In outbreaks, the birch skeletonizer *(Bucculatrix canadensisella)* completely destroys foliage by August. Successive attacks reduce ho attacks.

Although yellow birch is not a preferred host of the forest tent caterpillar *(Malacosoma disstria),* the gypsy moth *(Lymantria dispar),* looper *(Lambdina fiscellaria),* or the saddled prominent *(Heterocampa guttivitta),* caterpillars of these species defoliate birch in sever can kill birch trees (122). Dusky birch sawfly larvae *(Croesus latitarsus)* prefer small saplings of gray birch but also defoliate yellow b lace bug, *Corythucha pallipes,* can be a very injurious insect, especially on young birch (25). A treehopper *(Carynota stupida),* a stink a lygaeid bug *(Kleidocerys resedae germinatus),* and a scale insect (*Xylococculus betulae)* are other commonly to abundantly occu *Kleidocerys resedae germinatus* also feed on catkins. The birch seed midge *(Oligotrophus betheli)* lives in birch seed and makes it infe

Yellow birch is a preferred food of the snowshoe hare and the white-tailed deer. White-tails are especially fond of browsing seedlings o the fall, and they favor succulent sprouts over slower growing seedlings. Heavy or repeated browsing often kills seedlings. Moose of birch crowns, reduces wood quality, and sometimes kills the trees. Red squirrels cut new germinants, eat seeds, store mature strobiles

Yellow birch is a favorite summer food source of the yellow-bellied sapsucker on its nesting grounds. Heavy sapsucker feeding can common redpoll and many other songbirds eat yellow birch seed. Ruffed grouse feed on the catkins, seeds, and buds.

Special Uses

Yellow birch lumber and veneer are used in making furniture, paneling, plywood, cabinets, boxes, woodenware, handles, and interior d distillation of wood alcohol, acetate of lime, charcoal, tar, and oils (9).

Genetics

Population Differences

Yellow birch shows great phenotypic variability (17,28). Significant variation in catkin, fruit, and vegetative characteristics have been rep (15,30,102).

A range-wide study of 55 provenances indicated random variation in seedling height and diameter growth but clinal variation in grow grown from seed sources collected along latitudinal and elevational gradients in the Appalachian Mountains were tested, stage of lea no influence on stage of flushing and time of growth cessation was more closely related to latitude than elevation (101). Thus, season in yellow birch appear to be under strong genetic control and probably evolved as an adaptation to different photoperiods (16). Temp breaking of dormancy and cessation of growth (17,101), and provenances differ in their chilling requirements. Even though the maxim by geographic seed source (44), overall hardiness apparently does. Seedlings from southern seed sources and low elevations su northern and eastern sources (24).

After 5 years, northern seed sources survived better than southern sources in three northern field plantings, but height growth was not indicate the importance of collecting seed for planting from superior local phenotypes growing on sites similar to the intended planting s

Within- population variability in morphology, phenology, and growth of yellow birch is large and often exceeds among-population variat fruit characteristics (15), bark color and exfoliation (28), and periodicity of shoot growth (18,101). After 5 years in the field, yellow birch diameter, and crown size (23).

Races and Hybrids

Yellow birch trees with fruiting bracts 8 to 13 mm (0.3 to 0.5 in) long have been called var. *macrolepis* (Fern.) Brayshaw. However, beca throughout the natural range of yellow birch (15,29,102), it should not be considered a separate variety. Variety *fallax* (Fassett) Braysha exfoliate in thin shreds as much typical yellow birch bark and resembles bark of nonexfoliating *Betula lenta L.* This

dark-barked form occurs in southern Michigan, southern Minnesota, and southern and central Wisconsin, northern Indiana, and northe

Yellow birch hybridizes naturally with low birch *(Betula pumila* L. var. *glandulifera* Reg.) This hybrid *(B. x purpusii* Schneider) can b length, blade width, stomatal length, leaf apical angle, number of teeth per side of leaf blade, and pollen diameter (29). Murray birc Michigan. It appears to be an octoploid derivative of an unreduced gamete of Purpus birch and a reduced gamete of yellow birch. Mur supposed ancestors and larger leaves and fruits than Purpus birch (4). Mountain paper birch *(B. papyrifera* var. *cordifolia* (Regel) F birch *(B. papyrifera* Marsh.). F, hybrids are intermediate between their parents in most characteristics but they can be separated pubescence, bract lobe length, amount of bractcilia, and the number of leaves per short shoot (5). This hybrid grows better than yello At 27 years of age, *B. lenta x B. alleghaniensis* hybrids were about 0.9 m (3 ft) shorter in height and 23 mm (0.9 in) smaller in diameter

Through controlled pollination the interspecific hybridity of yellow birch and the following species has been verified: *B. papyrifera* M *pendula* Roth., *B. davurica* Pall., *B. nigra* L., *B. occidentalis* Hook., *B. populifolia* Marsh., *B. mandshurica* (Reg.) Nakai, and *B. pubesce*

B. alleghaniensis is a hexaploid with normal vegetative cells having 42 pairs of chromosomes. Although some counts as high as 49 ha counts of 42 with a range between 37 and 49 pairs (17).

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